

# API alignment report

Next-Generation IEDB Tools API · NIAID Blueprint Section 3 · generated from niaid-bp-api-assess

## Input

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- **Source:** <https://api-nextgen-tools.iedb.org/apidocs>
- **Kind:** openapi (the docs URL itself parsed; hop: none)
- **Spec format:** Swagger 2.0 — *Next-Generation IEDB Tools API*, info version v3.0, server <https://api-nextgen-tools.iedb.org>
- **Confidence:** high. A spec parsed and response media types are present. No samples were attempted.
- **Outside this rubric:** none

The document describes 42 operations. They are tool and job endpoints (MHC prediction, peptide tools, pipelines, plots, sequence downloads), not a dataset catalog. Every operation is marked secured, so no unauthenticated sample was planned.

## Blueprint alignment

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Section 3 asks for JSON-LD (at least as an option), resource IRIs usable as @id, GET for retrieval, and an OpenAPI/Swagger document.

### bp\_jsonld\_declared — High

- **Evidence:** No response offers application/ld+json, and no schema or example declares @context. Documented media types are application/json (for example GET /api/v1/results/{result\_id}), text (GET /api/v1/download\_sequences/{sequences\_id}), or application/json plus text/fas (GET /api/v1/sequences/{sequences\_id}). That pattern holds for all 42 operations.
- **Recommended next step:** Add application/ld+json (or a ?format=json-ld response) on at least the resource reads — GET /api/v1/results/{result\_id}, GET /api/v1/pipeline/{pipeline\_id}, GET /api/v1/sequences/{sequences\_id} — with @context and @id using schema.org types from Blueprint Table 1.

### bp\_resource\_iris — Medium

- **Evidence:** GET /api/v1/alleles/search (verb-like segment search); GET /api/v1/get\_object/{object\_id} (verb-like segment get\_object). Other paths are noun-shaped (/results/{result\_id}, /sequences/{sequences\_id}).
- **Recommended next step:** Replace those two with resource IRIs, for example /alleles/{allele\_id}, and /objects/{object\_id} instead of get\_object.

### bp\_version\_in\_path — Medium

- **Evidence:** Segment v1 is in the path of every /api/v1... operation, including GET /api/v1 and GET /api/v1/results/{result\_id}. The info version is v3.0 while the paths say v1. A version segment is unstable if that URL is used as a JSON-LD @id.
- **Recommended next step:** Keep version negotiation in a header or a media-type parameter, and publish stable IRIs without /v1/ for any object that should be an @id.

**Passed:** bp\_openapi\_doc (Swagger 2.0 parsed), bp\_get\_retrieval (retrieval is GET; the POSTs are not bare retrievals).

**Not stated (not a failure):** bp\_jsonld\_observed. No unauthenticated sample was fetched, so this run does not claim the live API returns or lacks JSON-LD.

## Common practice

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These are not Blueprint failures.

### prac\_info\_license

- **Evidence:** info.license name or url is absent.
- **Recommended next step:** Add a license name and URL on the Swagger info object.

### prac\_operation\_summaries

- **Evidence:** No summary or description on 12 operations, including GET /api/, GET /api/v1/alleles/search, GET /api/v1/citations, GET /api/v1/pipeline/{pipeline\_id}, GET /api/v1/pipeline\_spec/{pipeline\_spec\_id}, GET /api/v1/stage/{stage\_id}, and GET /api/v1/version\_info.
- **Recommended next step:** Add a one-line summary to each of those 12 operations.

### prac\_response\_examples

- **Evidence:** Every 2xx response has no example (all 42 operations).
- **Recommended next step:** Add one example body for GET /api/v1/results/{result\_id} and GET /api/v1/pipeline/{pipeline\_id} first, then the collection GETs.

### prac\_error\_responses

- **Evidence:** No 4xx or 5xx response is documented on any operation.
- **Recommended next step:** Document 401/404 (and 400 for the pipeline POST) once at the shared response level and reference them from each operation.

### prac\_pagination

- **Evidence:** Collection GETs have no limit, offset, page, or cursor. Examples: GET /api/v1/citations, GET /api/v1/mhci, GET /api/v1/pipeline/valid\_steps, GET /api/v1/results/{result\_id}/plots.

- **Recommended next step:** Document a limit and offset (or cursor) on each collection GET, or state in the operation description that the list is complete and unpaginated.

**Passed:** `prac_info_contact`, `prac_operation_ids`, `prac_local_refs`.

## Samples

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`sample_plan` was empty. Every operation is secured, and the script does not invent path ids or send credentials. No live GET was made. This run does not say whether any response body contains `@context`.

## Not checked

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- HTML pages with embedded JSON-LD, and a downloadable metadata index. Both are Blueprint fallbacks. This run did not crawl the site for them.
- Authenticated operations. No API key was sent.
- Table 1 fields that are absent from `observed_property_names`. Names that did appear: `cancel_run`, `email`, `input_parameters`, `input_sequence_text`, `pipeline_id`, `pipeline_title`, `run_stage_range`, `stage_number`, `stages`, `tool_group`. None of those is a Blueprint Table 1 dataset property.

## Overall

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The API is machine-documented (Swagger 2.0) and retrieval uses GET, so two Section 3 objectives are met. The highest-priority gap is metadata encoding: responses are declared as JSON or text, not JSON-LD, so a catalog cannot treat these URLs as linked metadata. Fix that before the path cleanup (`search`, `get_object`, and `/v1/`). The practice gaps (license, examples, errors, pagination) matter for clients, but they are not Blueprint misses. A landing-page JSON-LD check, if wanted, is `niaid-bp-metadata-extract` — this run did not look for HTML-embedded metadata.